Jairs

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A Jai-inspired systems language in Rust. Incremental salsa front end, lossless CST, typed SSA mid-end, two execution engines held byte-identical by a differential harness.

6/6 gates green **928 tests** **ADR-0068 latest** **W4.5 complete · W4 comptime next**

TESTS	CORPUS	ADRS	DIAGNOSTICS	EDITOR CHECKS
928	**155**	**68**	**91**	**166**
workspace, all passing	jr files, both engines	0001 to 0068, immutable	codes, E0261 next free	Neovim, verified not gated

The vertical slice, and what it costs to be honest about it

EXIT CRITERIA

- **DONE** hello.jr runs in the VM
- **DONE** compiles to a native binary
- **DONE** rustc-grade diagnostics
- **DONE** formatter round-trips the corpus
- **DONE** editor integration (Neovim)
- **OPEN** verified Linux x86-64 CI run

The last one needs a push. Configured, never run, so it is a decision rather than a technical gap.

HOW CORRECTNESS IS ESTABLISHED

Both engines run every corpus program and must agree, on output and on trap wording, from one shared formatter.

A plan's stated reason is checkable. W4.5 was scheduled after W4 because exhaustiveness “wants comptime type info”. Three greps and a two-line program showed it does not: the enum member set is already in the pool during checking. The wave moved forward and the table records the amendment — a wave order resting on a dependency that does not exist is a plan contradicting itself.

A new enum variant only asks the question; answering it is still manual. `AggregateKind` and `TagCheck` each turned every match site into a compile error, which is what **found** the sites — but a compile error only asks “which group does this belong to”, and this wave answered two of them wrongly. Both were silent: DCE deleted a variant's stores, and the slot-liveness collector panicked. Running found them; no verifier would have.

The formatter destroyed a declaration, again. A two-way `if` whose `else` meant “struct” turned every `variant` into a `struct` — the exact mistake that function's own docs already warned about for `enum_flags`, made again one form later. Thirteenth wave in fifteen.

The language today

WORKS, END TO END	ABSENT (WAVE)
Full integer tower, bool, string, pointers	pointer difference p - q, deferred
float32 and float64, plain IEEE-754, no traps	percent on floats, is_nan, math (W7)
struct, nominal, one level	
union, nominal, untagged — a cross-field read reinterprets	
variant — a tagged union: a wrong-case read traps, switch destructures	a recursive variant; eliding the check in an arm
enum and enum_flags, namespaced, bare dot-member, switch cases	an explicit backing type
Fixed arrays and views, bounds-checked, returnable	array literals, sub-slicing
cast and xx from context; operator overloading	unary, index and call overloading
Trapping arithmetic, wrapping variants, bitwise	transmute; float printing
if, else, while, for, break, continue, defer, using	
switch with exhaustiveness checking over an enum; else	patterns, ranges, guards; a jump table
Multiple returns, named args, literal defaults	#must; a multi-result call in a return
import, foreign, system_library, #scope_module	polymorphs, macros (W5)
One trivial compile-time run, folded	arbitrary run, RTTI, insert (W4)
Traps name their source line and the live call chain	a per-frame line; inlined frames (none exist)
Bounds checks, strippable by build setting or #no_abc	a per-index #no_abc; any other build setting
context, a hidden parameter passed by pointer	
Procedures as values: call, pass, return, struct field	a cross-file or foreign proc value; comparing one cast to a pointer
null, a context-typed pointer literal; malloc and free	
An allocator in the context, installed and called through	
Pointer offset p + n, n + p, p - n — element-scaled, unchecked	the difference p - q; p[n]; ordering
push_context, a block with its own copy of the context	
talloc / reset_temporary_storage — a per-context bump arena	hands out *u8 only; alignment; a growable region

No error-handling model yet — ADR-0008 reserves the slot and the first half exists (several return values); `#must` is owed its own ADR. **No GC and no RAII**, which is a design value rather than a gap. **A union is untagged**, so reading a field other than the one last written reinterprets bits: documented in three places because it cannot be diagnosed. **No indirect calls**, which is the single largest gap in the project and is what blocks the rest of W3.

Compiler internals, 17 crates

STAGE	STATE	NOTE
Lexer, parser, CST, typed AST	works	Hand-written, error-recovering, trivia-preserving
Formatter	works	Pure function over the CST; lost source in 7 of the last 8 waves
HIR, name resolution, modules	works	Flat import merge; cycles legal; export filtering
InternPool: types, values, layout	works	One layout computation and one integer evaluator, shared
Sema: signatures, checking	works	E0212 to E0257; no const-eval here, by design
MIR: typed SSA	works	Block parameters, not phis; explicit bounds check and zeroing
Mid-end	5 passes	Inline, forwarding, const-prop, DCE, plus the bounds-check strip
Const-eval	works	Runs MIR through the bytecode VM
VM: register bytecode, libffi	works	No JIT; indirect calls, and malloc from its own region
Cranelift back end	works	Aggregate returns via sret; indirect calls via func_addr
LLVM back end	not started	W8 owns it
Language server	12 caps	Diagnostics, hover, goto, completion, rename, actions, hints
Neovim integration	works	Runtimepath dir, no plugin manager; 166 checks
Driver	stub	Should consume the workspace notion that now exists

Roadmap: W1 and W2 closed, W3 started

W1 Data	done	Numeric tower, wrapping ops, enum, enum_flags, union, arrays, views, cast, xx, operator overloading. Closed by ADR-0048.
W2 Flow and scope	done	for with it and it_index, labelled break, defer, using, multiple returns, named and default arguments, #scope_module. Closed by ADR-0054; ADR-0055 then closed a gap six corpus files had carried for eleven waves.
W3 Runtime core	done	context (ADR-0057), the bounds-check build setting (ADR-0058, finishing ADR-0003), indirect calls (ADR-0059), null plus a memory source (ADR-0060/0061), the allocator protocol (ADR-0062), push_context (ADR-0063), pointer arithmetic (ADR-0064), temporary storage (ADR-0065), and traps with backtraces (ADR-0066) — a trap names the frames that were live, byte-identically in both engines. Closed by ADR-0066; a source-level backtrace with inlined frames is deferred, because inlined frames have no runtime existence.
W4 Comptime	not started	Arbitrary compile-time execution, RTTI and Type values, insert, code. PLAN section 5 names this the project's top risk: sema and comptime become mutually recursive.
W4.5 Pattern matching	done	switch with exhaustiveness checking, a bare dot-member as a case (settling ADR-0041 §2 step 5), and a tagged variant type (ADR-0067, ADR-0068). Reordered ahead of W4 after checking showed its stated dependency on comptime was a want rather than a need. The variant follows ADR-0045 §1's own instruction — a different declaration form, not a change to union — and union is untouched, still untagged and still one word smaller.
W5 Polymorphism	not started	Polymorphic parameters, modify, bake_arguments, expand macros with hygiene, instantiation caching.
W6 Metaprogram	not started	Workspaces, the compiler message loop, build scripts replacing makefiles, note attributes.
W7 Stdlib	not started	Written in Jairs: String, dynamic array, hash table, Sort, Math, Random, File.
W8 Performance	not started	LLVM via inkwell for release builds, inliner maturity, struct-of-arrays, SIMD, parallel sema. Also owns the compile-throughput number, still unpublished.
W9 Tooling depth	not started	Semantic tokens are the one LSP capability still missing; the other twelve landed early.
W10 Graphics, in Jairs	not started	Window creation through foreign calls, Metal then Vulkan, immediate-mode 2D.

What this session did, and how each wave was scoped

TWENTY WAVES SHIPPED

ADR-0049 through 0068: for and defer, using, aggregate returns, multiple returns, named and default arguments, scope visibility, imported constants, float constants, context, the bounds-check build setting, indirect calls, null plus a memory source, the allocator protocol, push_context, pointer arithmetic, temporary storage, trap backtraces, switch, and tagged variants.

Test count 900 to 928. Corpus 116 to 155 files. Neovim checks 103 to 166. **W2, W3 and W4.5 are all closed** — W4.5 a wave early, because its stated dependency on comptime turned out not to exist. A switch over an enum or a tagged variant is exhaustiveness-checked, and a wrong-case read traps. What remains is W4 — comptime, the project's top risk.

Each wave was scoped by writing the feature and seeing what the compiler refused, not by reading the handoff. That found gaps the handoff had missed in both directions: a proc-pointer struct field it called absent (it worked), and a void-returning proc-pointer type it never mentioned (unspellable, and blocking).

TWO CLAIMS THE CODE DISPROVED

An ADR corrected mid-wave. ADR-0060 §4 asserted the VM dereferences a `malloc`'d pointer via libffi. Running the corpus file the same ADR promised would pass disproved it: the VM's memory is a linear region and a host address is not an offset into it. ADR-0061 corrects it — the VM allocates from its own region, native calls libc, the bits differ and nothing observes them.

A diagnostic that could not be acted on. Installing an imported `#foreign` procedure into an allocator field reported "expected `(s64) -> *u8`, found `(s64) -> *u8`" — the same text twice, because the types differ only in an invisible calling convention. It is E0256 now, the code that says to wrap it.

THE THING WORTH YOUR DECISION

Fourteen waves sit uncommitted across five branches. Two waves ago a careless `git checkout` reverted the tree-sitter grammar nine waves and cost an hour's reconstruction — the concrete price of not committing. Committing each wave as it goes green would bound the damage from any future slip to a single wave.

Recorded as a recommendation, not a change: the authorisation is yours to give.

Sources: PLAN.md §1.5 and §7, the ADR directory, `docs/decisions/DECISIONS.md`, and the six gates run today. Every number was measured rather than carried forward — the test count from a full workspace run, the corpus count from a file walk, the diagnostic codes from the `const E0nnn` definitions across the crates, and the editor-check count from a real `verify.lua` run. Rebuild with `typst compile jairs-dashboard.typ`.